

GABLS1: measured scheme-native WENO flux

GABLS1 scheme-native WENO transport | 22 September 2026. Matched 12.5 m, 32³, 400 m, WENO9, one face, 300 s filter, factor one, seed 123, nine-hour LES. The old admitted factor-one covariance run is the control.

At the first interior face, the measured 8–9 h WENO numerical correction to u-momentum flux is $-0.000228 \text{ m}^2/\text{s}^2$, versus covariance $-0.007749 \text{ m}^2/\text{s}^2$ (2.95%). The heat correction is $1.4\text{e-}10 \text{ K m/s}$, effectively zero versus covariance -0.001472 K m/s . The saved covariance plus correction equals reconstructed flux at both diagnostic faces for momentum and scalars. These measured corrections are far below the assumed extra flux behind factors two and three in this case.

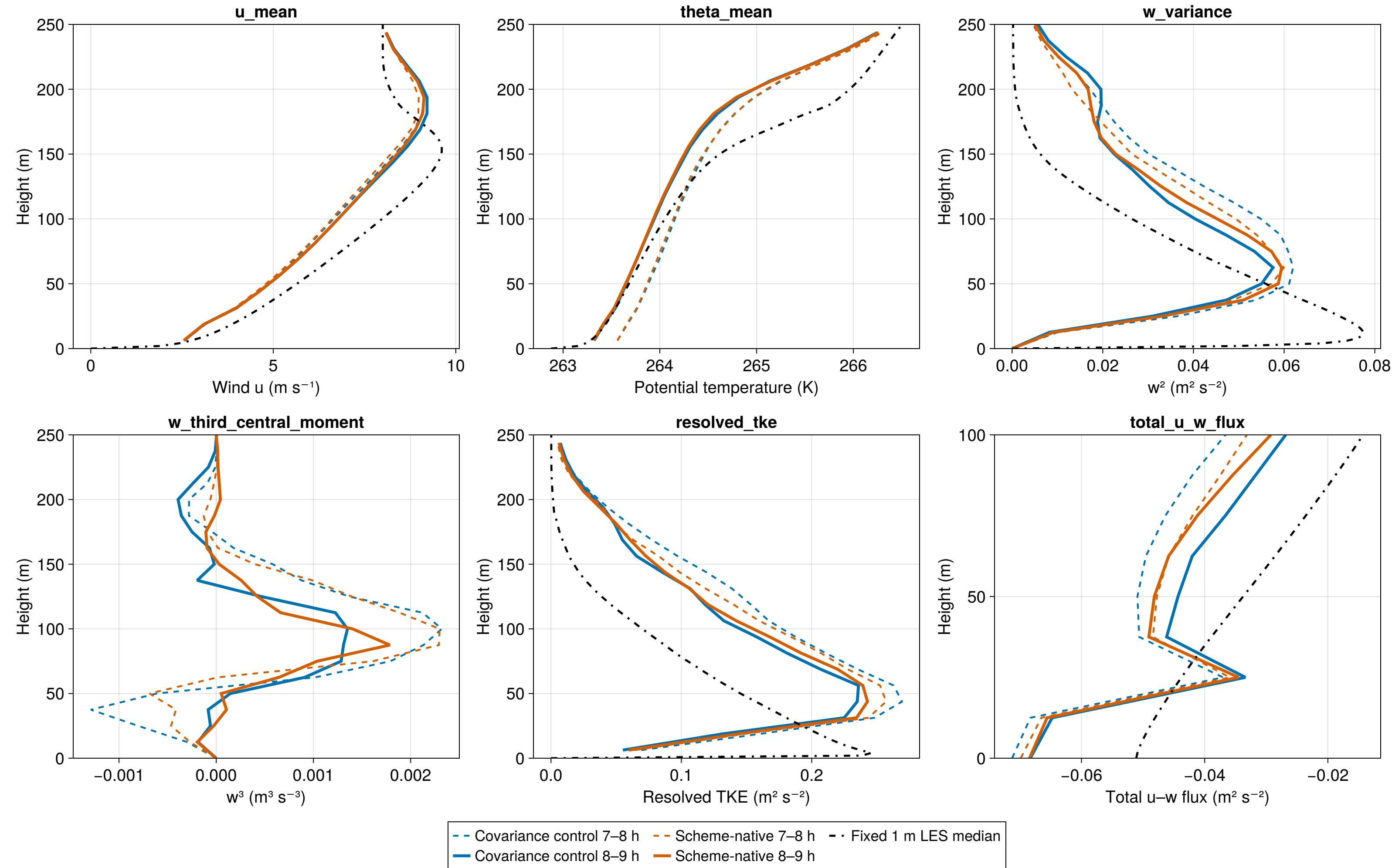
Mean 8–9 h first-face momentum viscosity falls 1.3214 to 1.2958 m^2/s (-1.9%); heat diffusivity rises 1.2160 to 1.2642 m^2/s (+4.0%). Integrated resolved TKE rises 3.8%, peak w^2 rises 3.1%, diagnosed boundary-layer height falls 7.4 m. One evolving realization cannot distinguish a systematic response from trajectory divergence. The profiles remain near the covariance control and show substantial coarse-run departures from the fixed 1 m LES median in resolved turbulence.

The final-hour mean horizontal-wind vector shear across 6.25-18.75 m is 0.0522 s^{-1} for scheme-native SLD and 0.0511 s^{-1} for covariance SLD, versus 0.1004 s^{-1} from the fixed 1 m median profiles; the no-closure case is high at 0.1861 s^{-1} . The pattern persists during 7-8 h, but differences change sign higher up. All model gradients use adjacent 12.5 m cell-center levels. Reference componentwise median u/v profiles are sampled at those heights and differentiated identically; this is not the median of member shears or a wall gradient.

Source and data audit: immutable Breeze/Evaluation manifest SHA-256 72f1bc03 and 792 source-file hashes passed; paired initial-theta SHA-256 1f5db33f matched. H100 gate 7496 passed; science 7497 had durable exit zero and CASE_DONE at 32400 s. Raw output passed exact 1/19/541 initial/profile/series schedules, finite values, native 32/33-level heights, and flux decomposition on both faces. Figures use Julia. The reference is an LES median, not observations. Historical factor and control outputs were not rerun.

GABLS1: scheme-native transport versus covariance

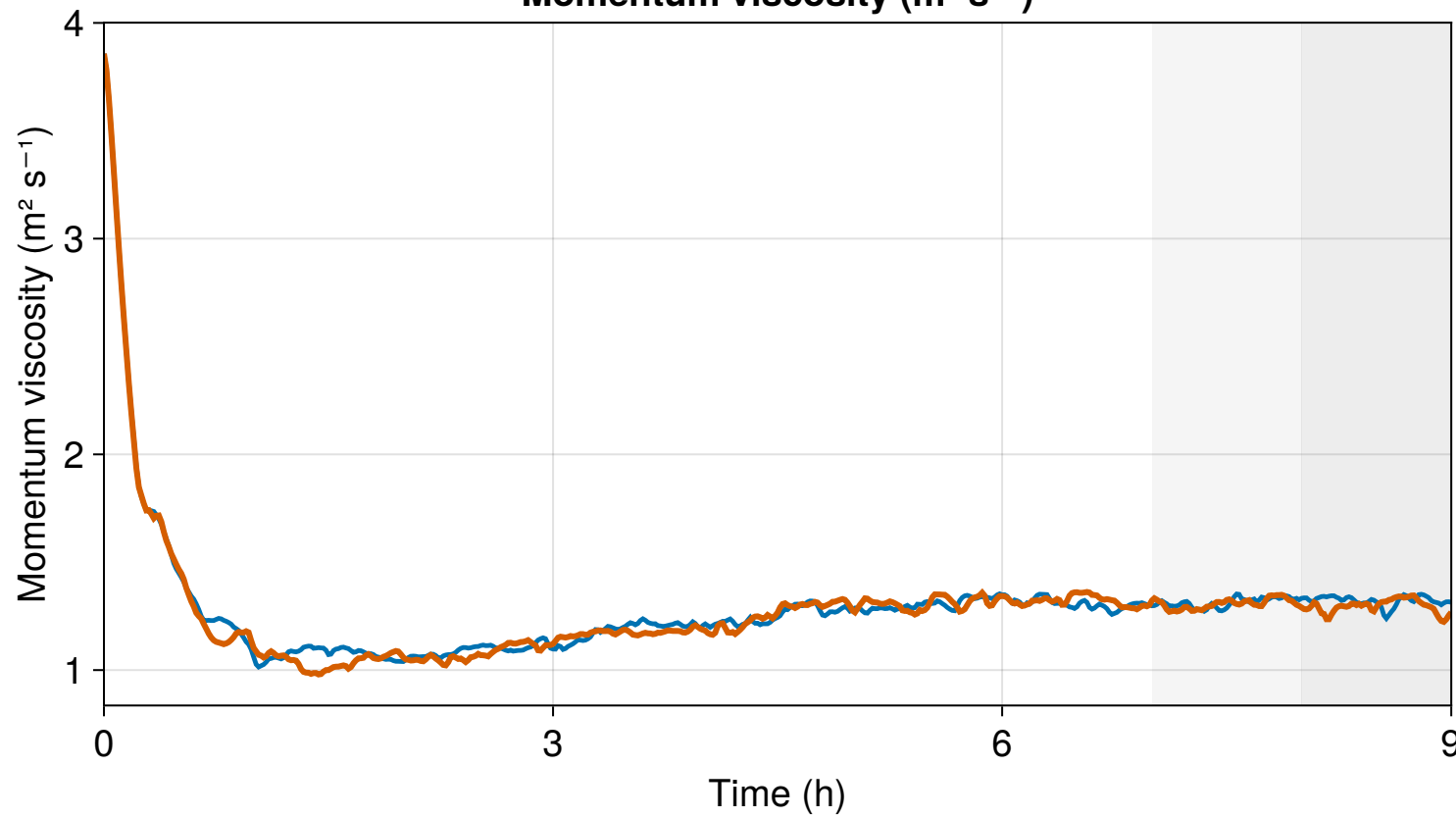
12.5 m · WENO9 · one face · 300 s · factor 1 · same seed; dashed 7–8 h, solid 8–9 h



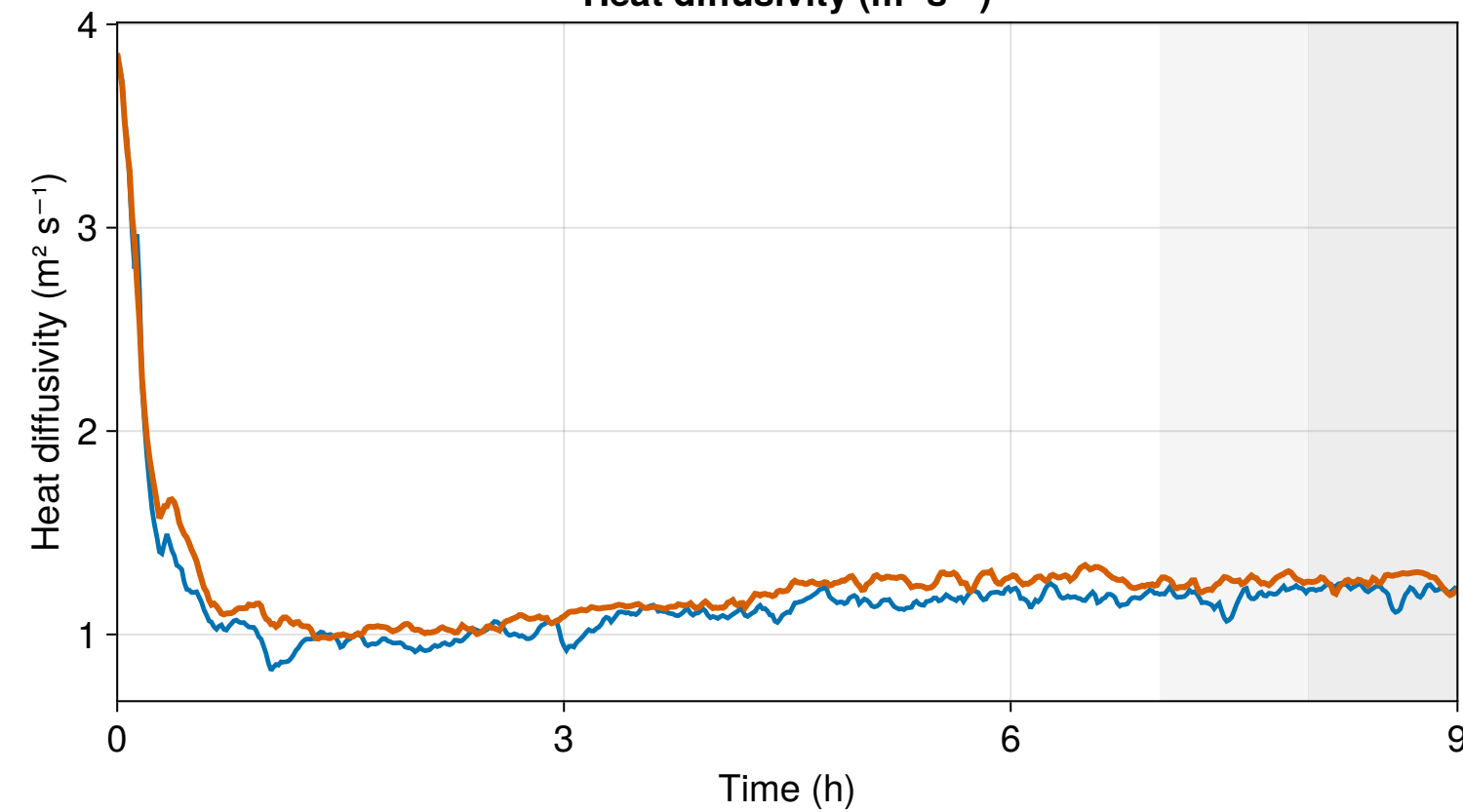
What the transport estimator changes at the first interior face

Same nine-hour forcing and discretization · 12.5 m supported face · shaded final two hours

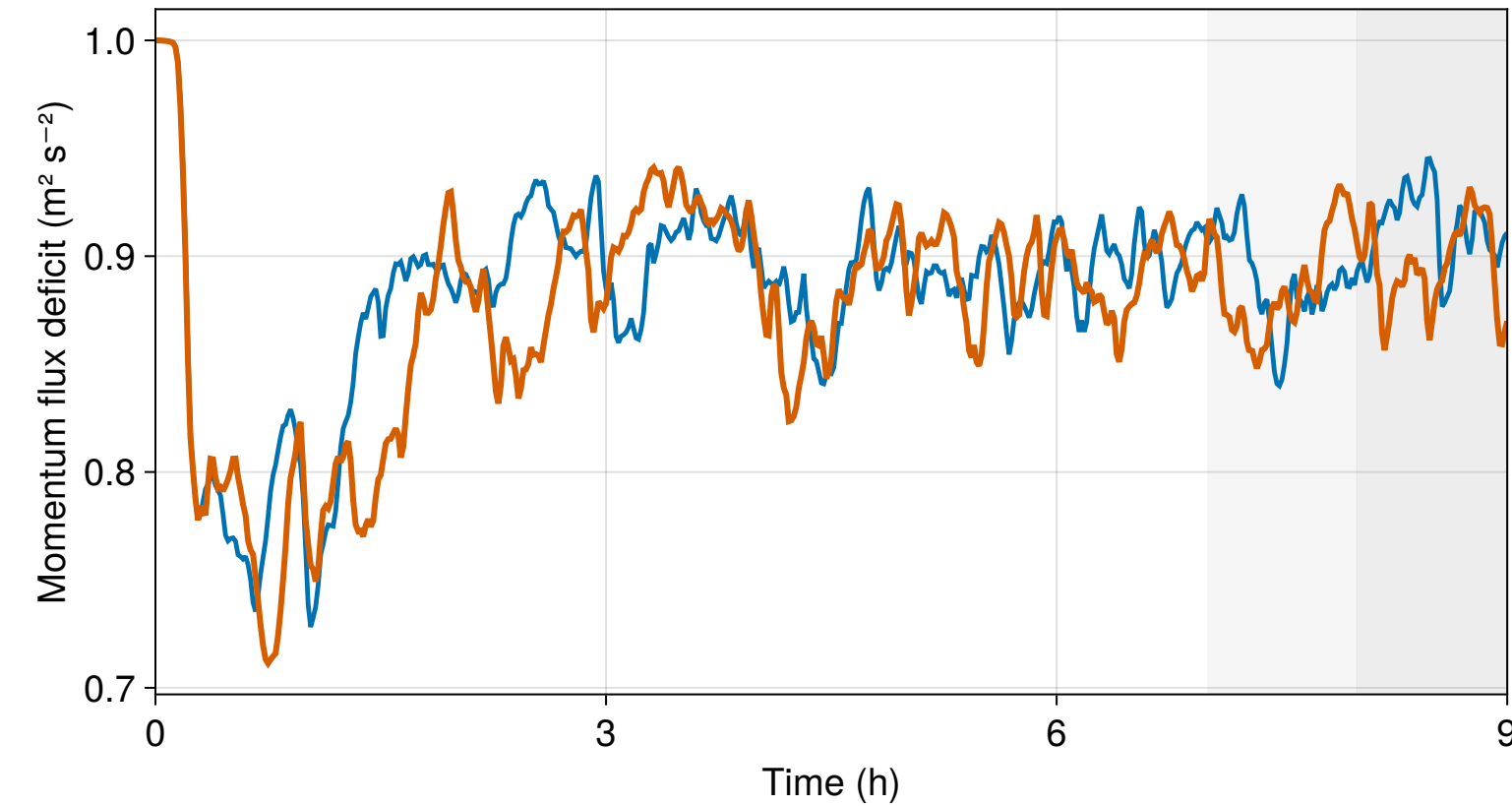
Momentum viscosity ($\text{m}^2 \text{s}^{-1}$)



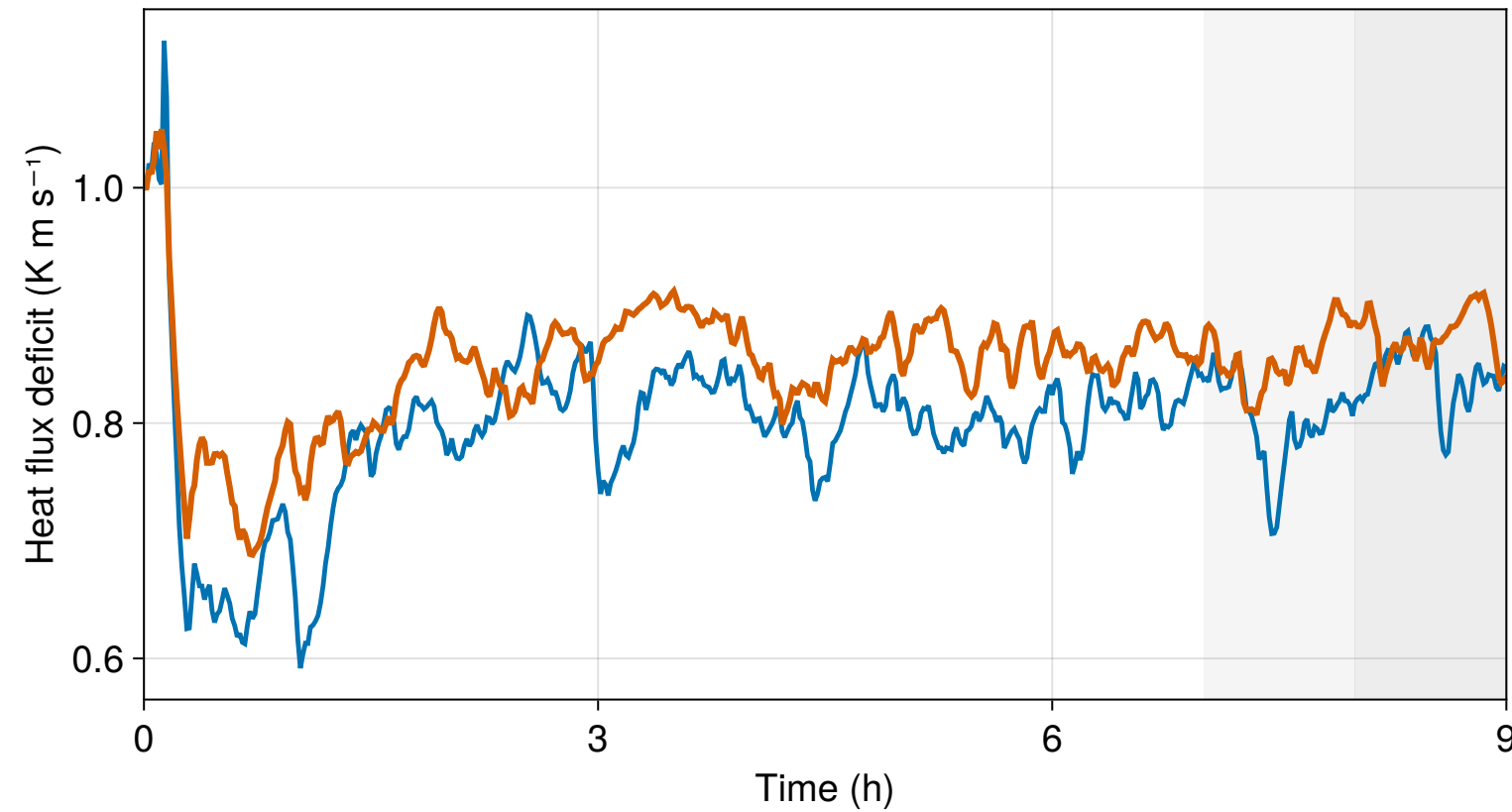
Heat diffusivity ($\text{m}^2 \text{s}^{-1}$)



Momentum flux deficit ($\text{m}^2 \text{s}^{-2}$)



Heat flux deficit (K m s^{-1})

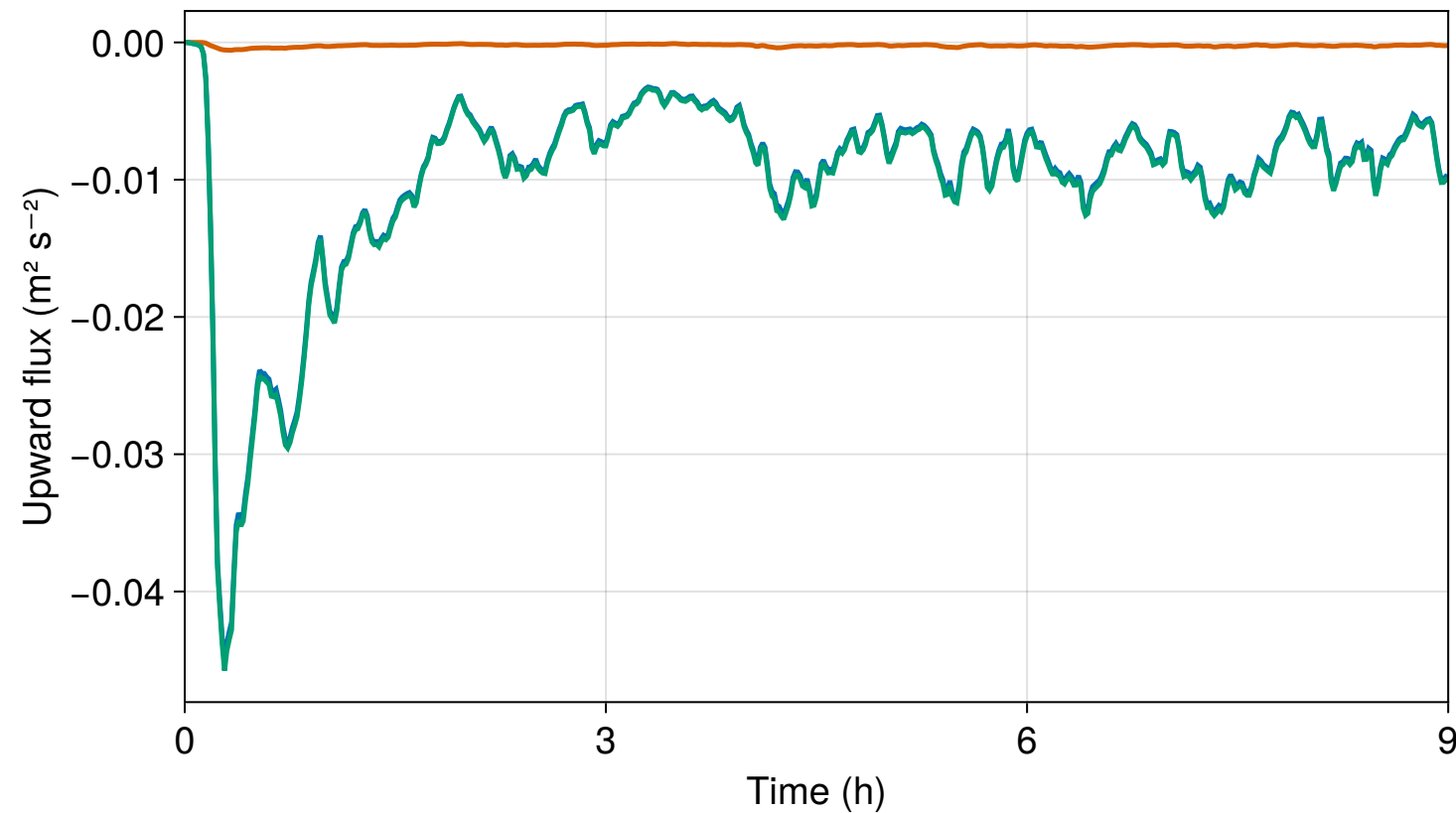


— Covariance control — Scheme-native

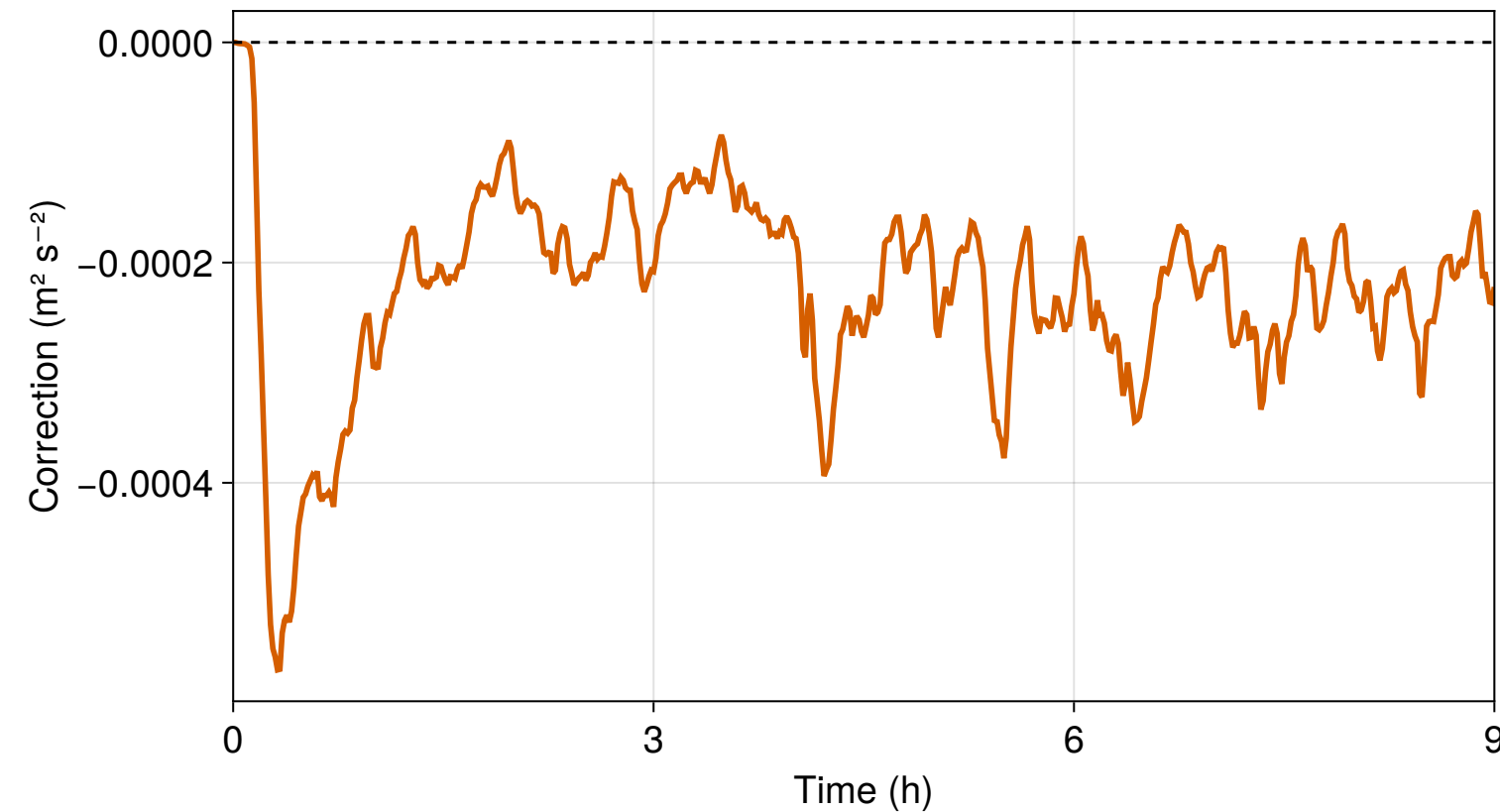
Measured WENO transport correction

Scheme-native run only · covariance + numerical correction = reconstructed flux

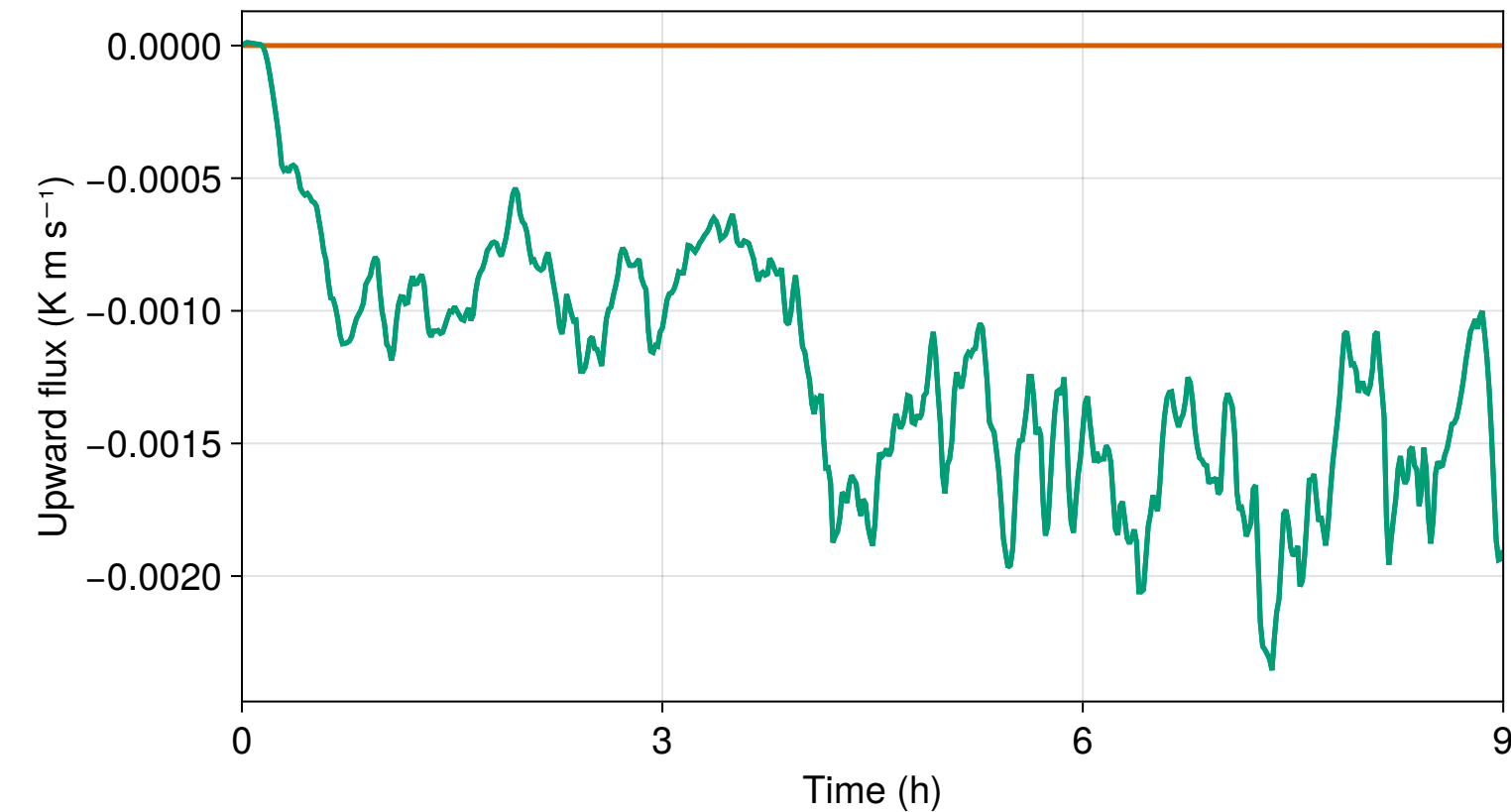
Momentum flux



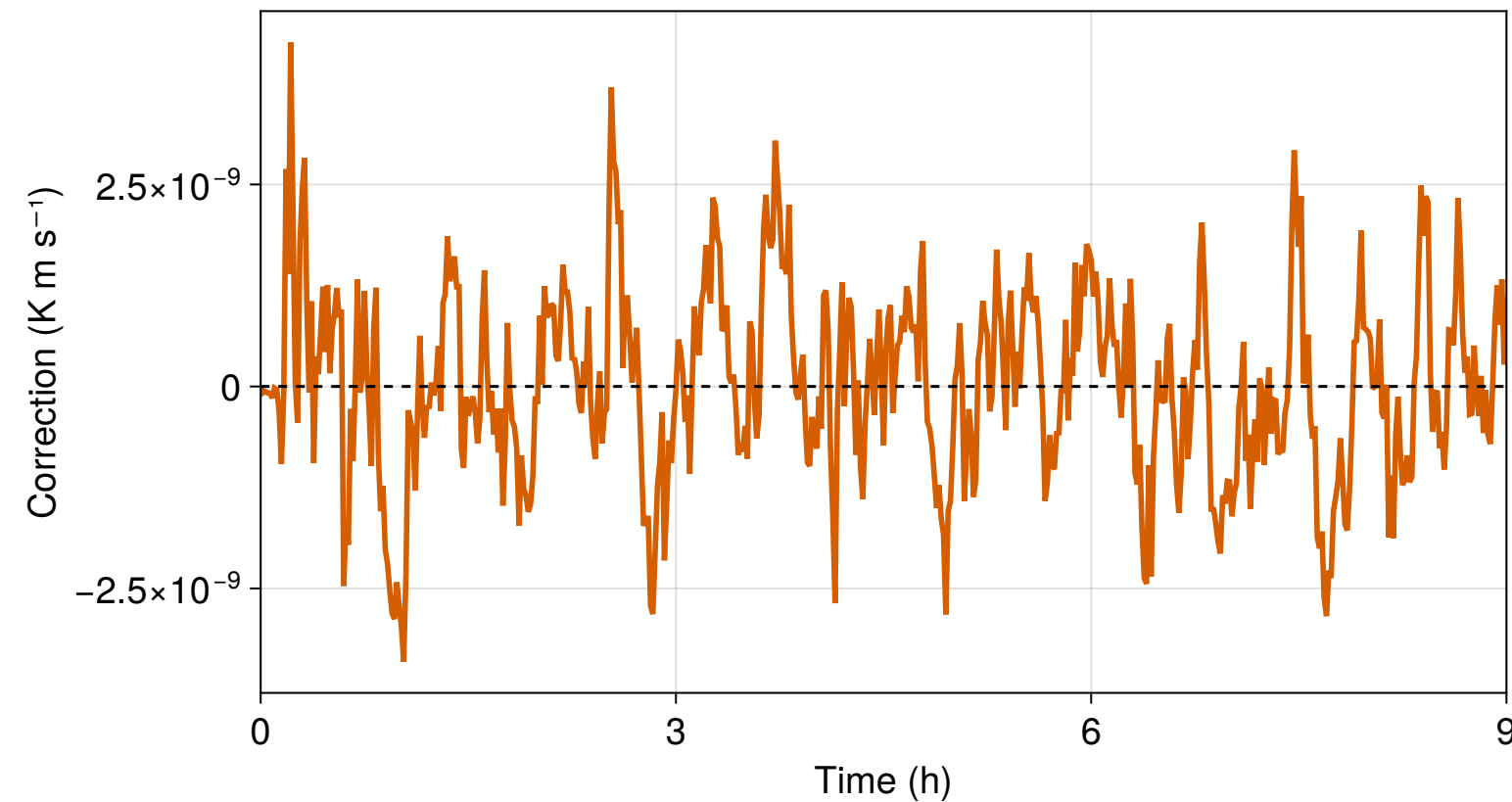
Momentum numerical correction



Heat flux



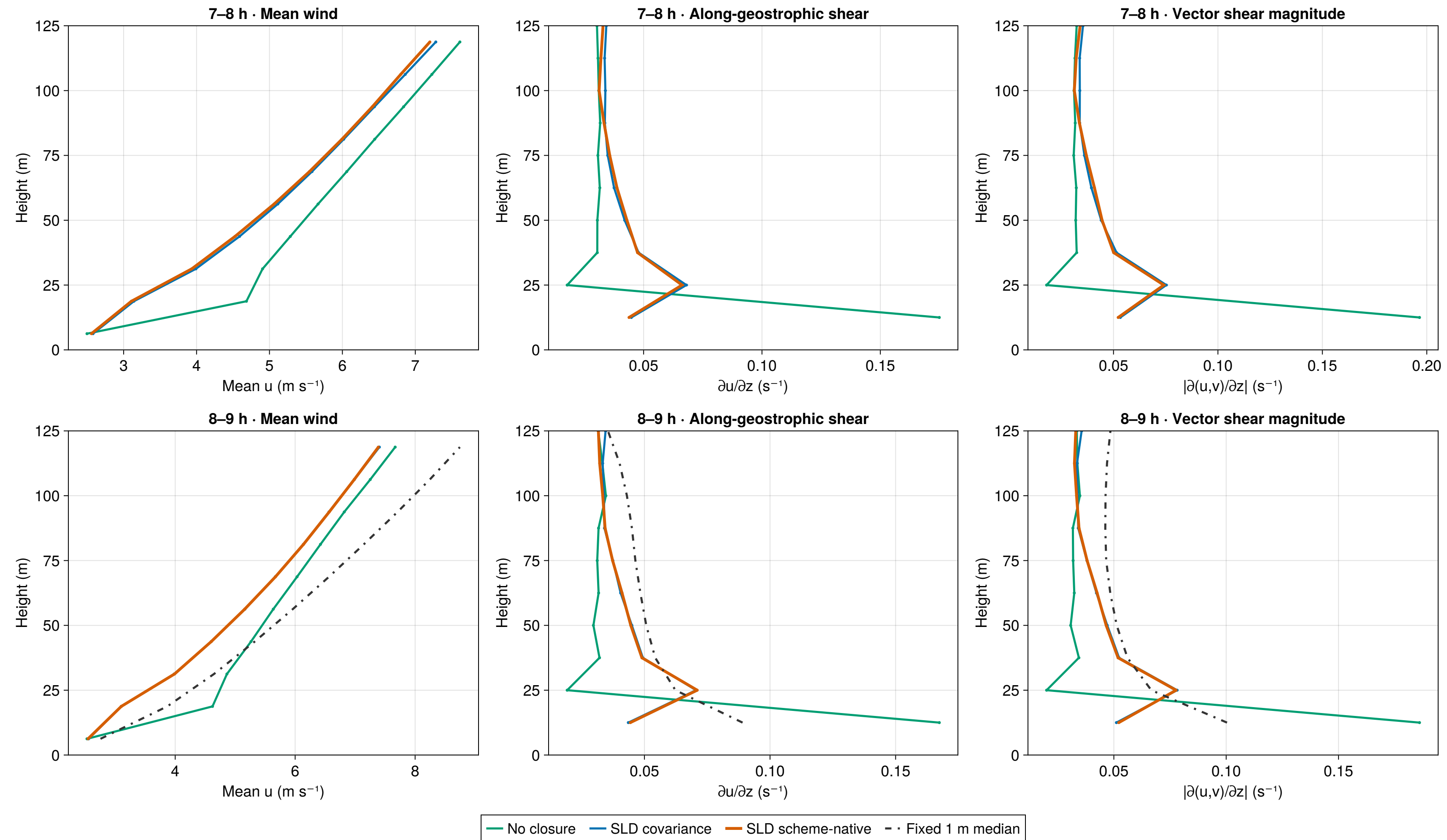
Heat numerical correction



— Covariance — Numerical correction — Reconstructed

GABLS1: vertical shear of the mean horizontal wind

12.5 m WENO9 · same 400 m cube · derivatives between adjacent cell-center heights



Reference is for 8–9 h only: componentwise median u and v profiles sampled at 12.5 m cell centers, then differentiated identically.
The reference shear is not the median of member shears. Each curve is a layer slope at 12.5, 25, 37.5... m, not a wall gradient.